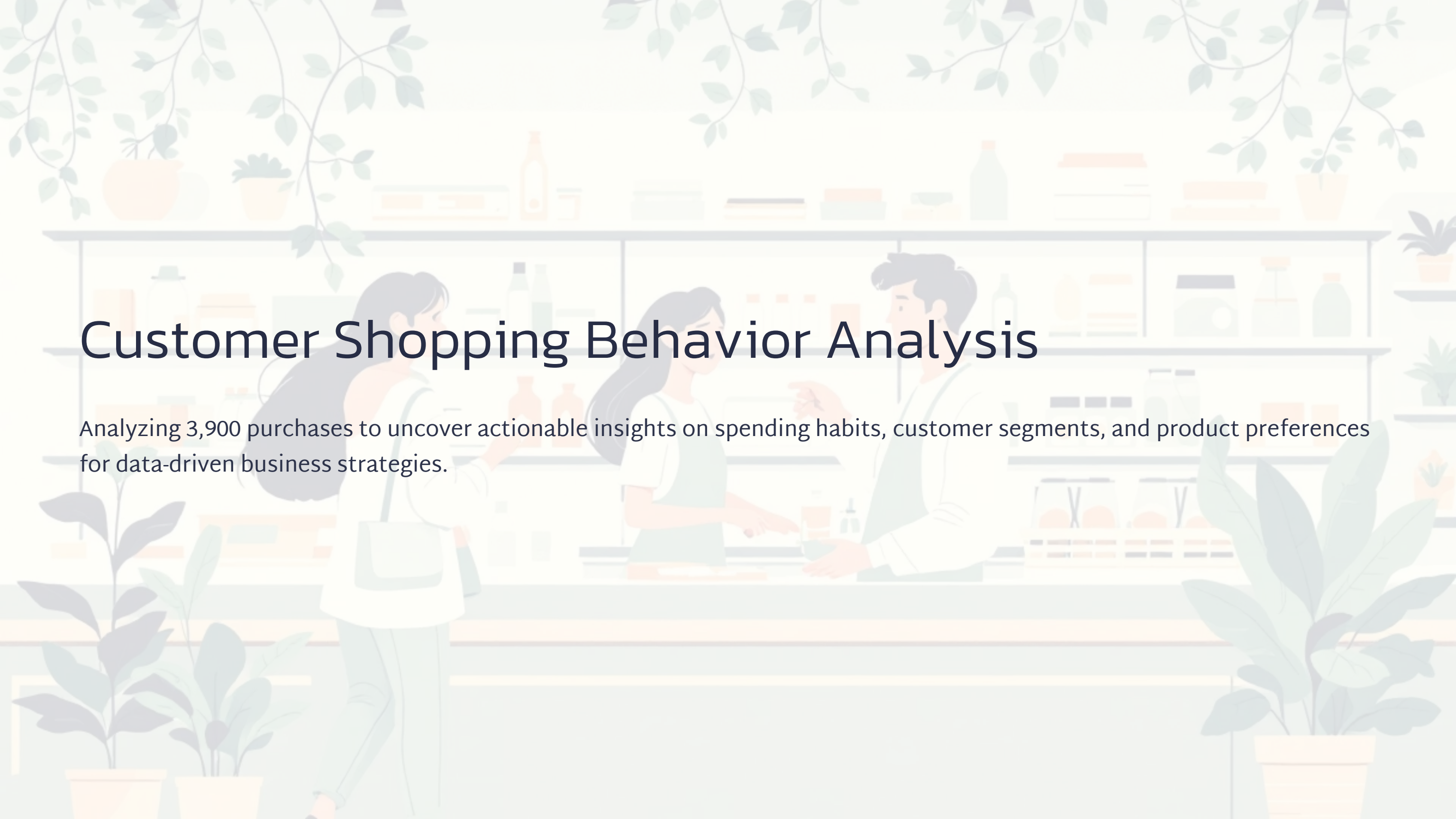




Customer Shopping Behavior Analysis

Analyzing 3,900 purchases to uncover actionable insights on spending habits, customer segments, and product preferences for data-driven business strategies.



Dataset Overview

3,900

Total Purchases

Transactions analyzed across diverse product categories

18

Data Dimensions

Features covering demographics, products, and transactions

\$59.76

Average Purchase

Mean transaction value in USD

44

Average Age

Mean customer age across all segments

Data Preparation Journey

01

Data Quality Check

Identified 37 missing values in Review Rating column requiring attention

03

Feature Engineering

Created age groups and numeric purchase frequency mappings for deeper analysis

02

Cleaning & Imputation

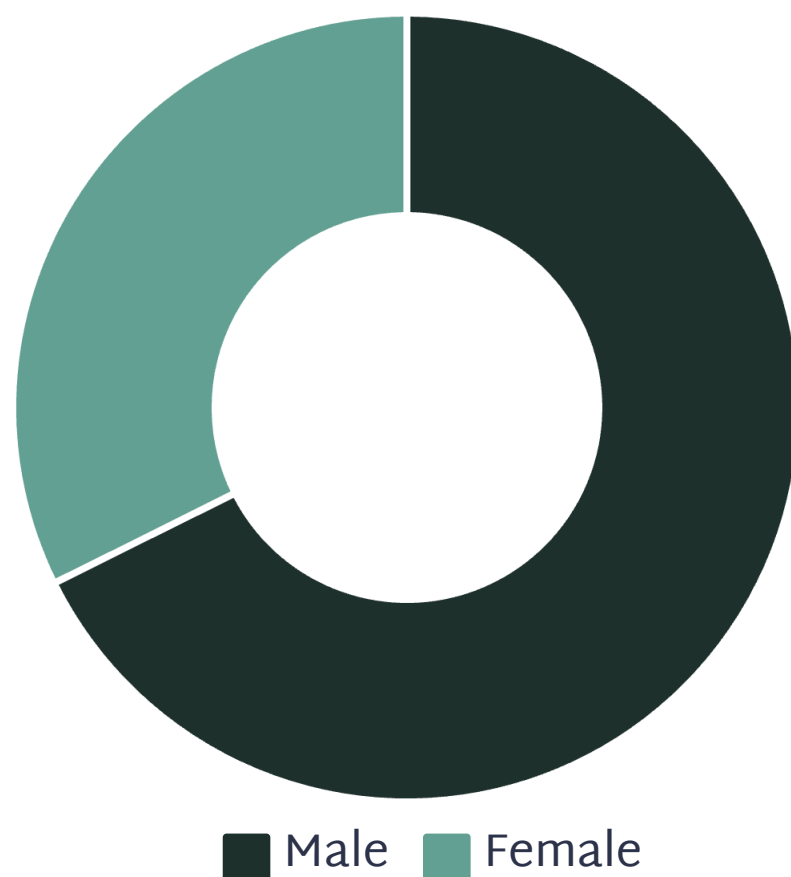
Filled missing ratings with category-specific medians to preserve data integrity

04

Database Publishing

Exported cleaned dataset to MySQL for advanced business intelligence queries

Revenue by Gender

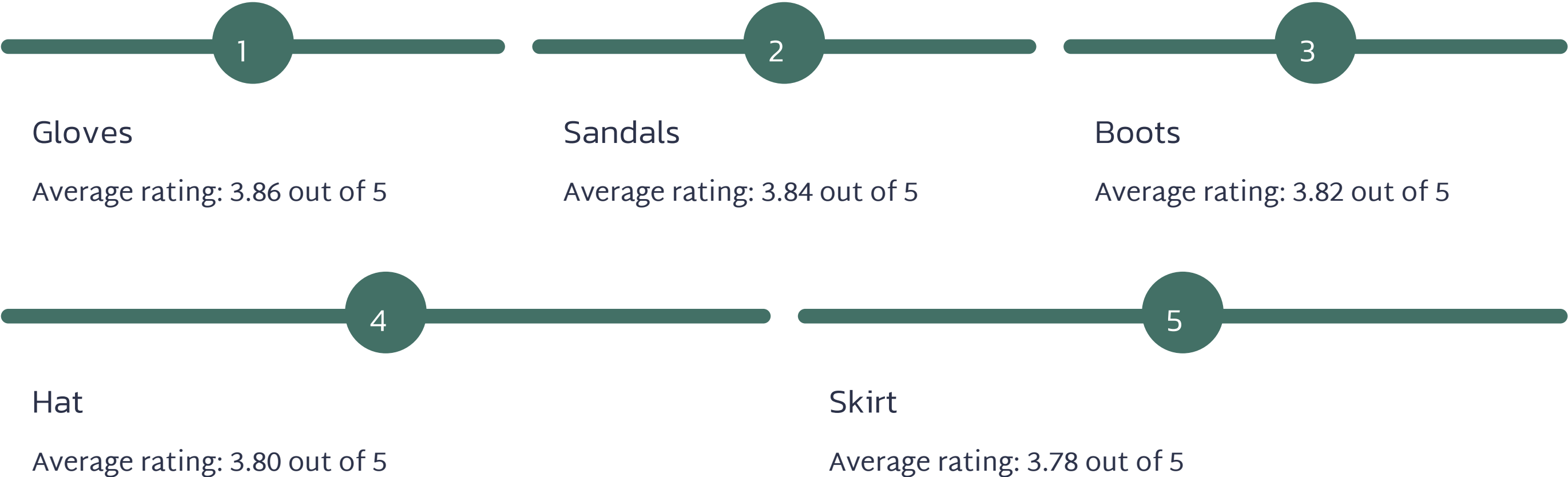


Male Customers Drive Revenue

Male buyers contribute more than double the revenue compared to female buyers, generating \$157,890 versus \$75,191.

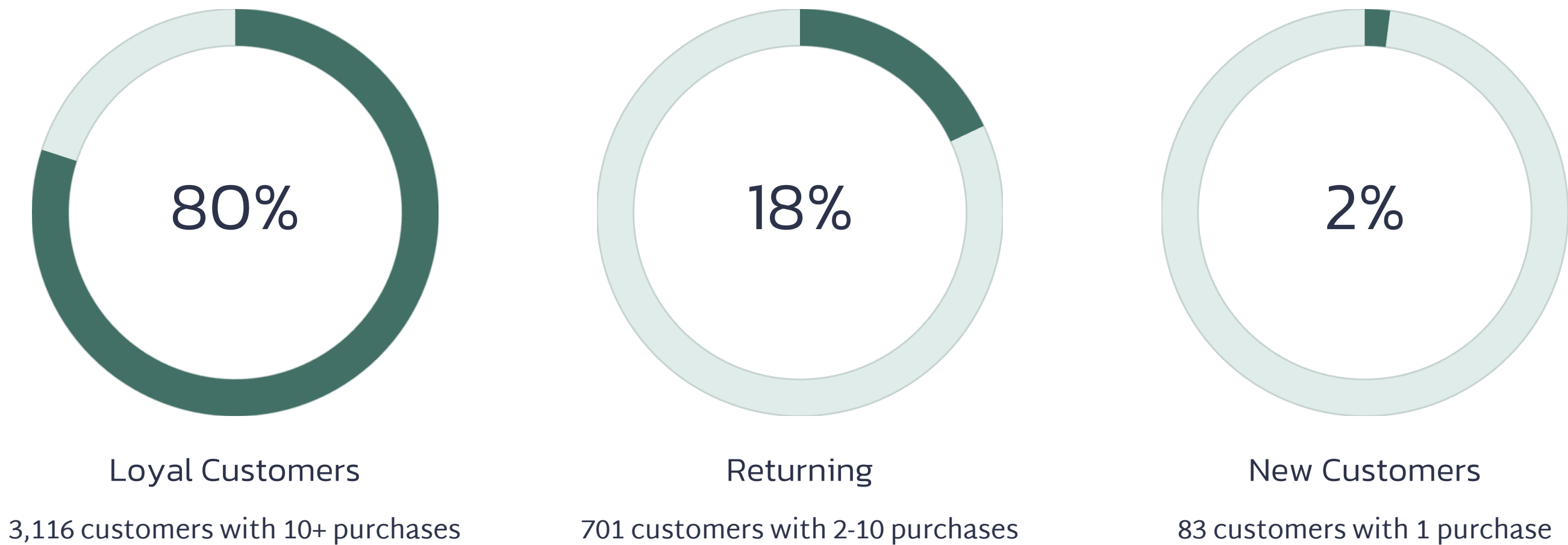
This significant gap presents opportunities for targeted marketing strategies.

Top-Rated Products



Customer satisfaction is highest in footwear and accessories categories, demonstrating strong product appeal.

Customer Segmentation Insights



Strong Retention Success

The large proportion of loyal customers indicates excellent retention and positive customer experience. Focus should shift to converting returning customers into loyal advocates.

Subscription Performance Gap

Subscribed Customers

Average spend: **\$59.49**

Total revenue: **\$62,645**

Among repeat buyers (5+ purchases): **958 subscribed**

Unsubscribed Customers

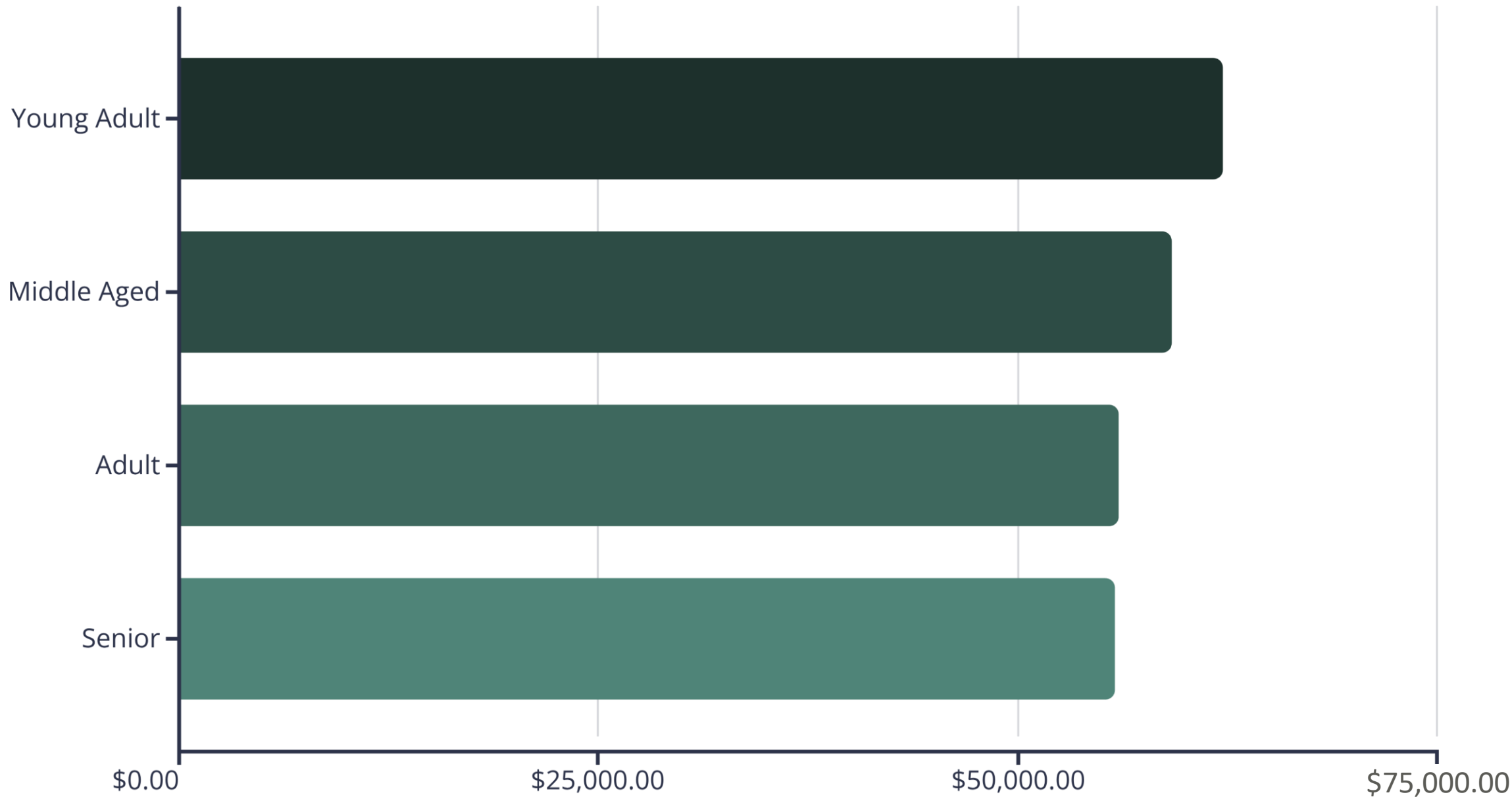
Average spend: **\$59.87**

Total revenue: **\$170,436**

Among repeat buyers (5+ purchases): **2,518 unsubscribed**

❏ **Critical Finding:** Unsubscribed customers spend more and generate higher total sales. The subscription program is not effectively influencing spending behavior and needs enhancement.

Revenue by Age Group



Young adults are the largest revenue contributors at \$62,143, forming the core customer base and primary driver of sales performance.

Dashboard Highlights



Customer Base

4,000 customers with \$59.76 average purchase value and 3.75 review rating



Category Leadership

Clothing leads in both customer engagement and revenue, followed by accessories and footwear



Untapped Potential

Most customers not subscribed nor using discounts—significant opportunity for loyalty strategies

Customer Behavior Analysis

Subscription

No

Yes

Gender

Female

Male

Category

- ☐ Accessories
- ☐ Clothing
- ☐ Footwear
- ☐ Outerwear

Shipping Type

- ☐ 2-Day Shipping
- ☐ Express
- ☐ Free Shipping
- ☐ Next Day Air
- ☐ Standard
- ☐ Store Pickup

Number Of Customers

4K

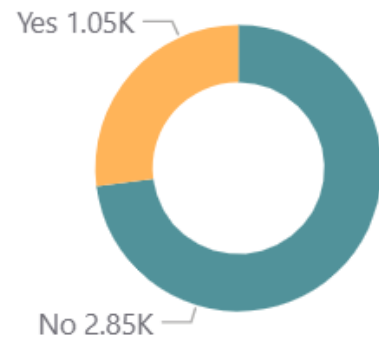
Average Review Rating

3.75

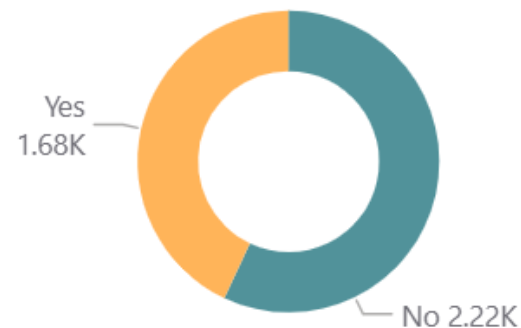
Average Purchase Amount

\$59.76

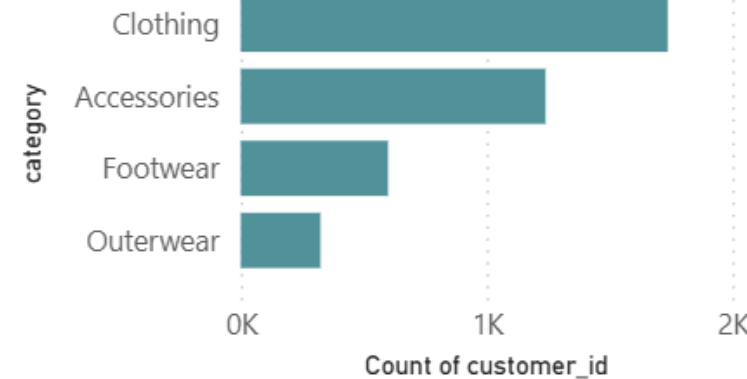
Customers By Subscription Status



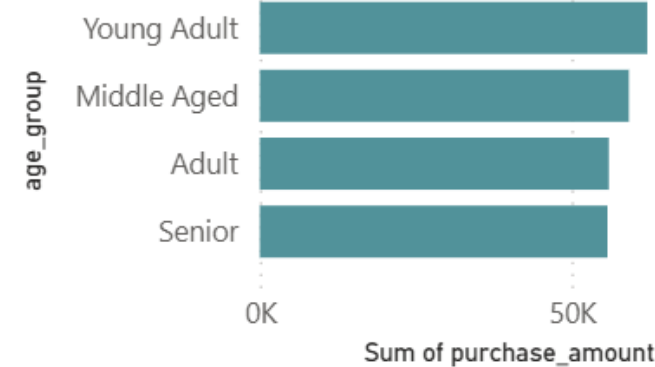
Customers With Discount



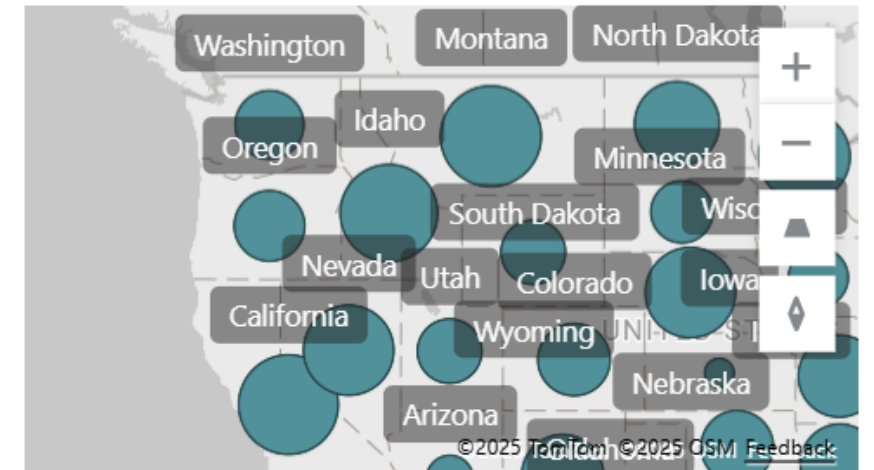
No of Customers By Category



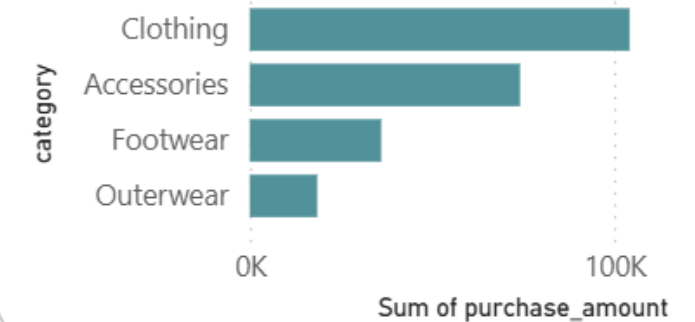
Revenue By Age Group



No of Customers By Location



Revenue By Category



Strategic Recommendations



Target Young Adults

Focus marketing on young adults using social media and influencer channels to maximize revenue impact



Expand Discount Offers

Offer strategic discounts and bundle deals on popular apparel items like hats and sneakers



Nurture Loyal Customers

Reward 3,116 loyal customers with personalized programs to maintain retention excellence



Revise Subscription Strategy

Add compelling perks to boost subscription sign-ups and spending among repeat buyers



Enhance Top Products

Prioritize inventory investment in highly rated items like gloves, sandals, and boots



Regional Marketing

Deploy localized campaigns in high-density customer areas for targeted outreach



Conclusion

In conclusion, these insights offer a strategic framework for driving sustainable growth in the retail sector. By leveraging targeted marketing strategies, enhancing customer retention initiatives, and implementing effective product management practices, businesses can establish a competitive advantage and achieve long-term success in a dynamic retail market.